PHILOSOPHY 1002: Introduction to Critical Thinking
WORKSHOPS for Part 1

Workshops in this part of the unit usually concern the material covered immediately prior to the first workshop of the week. For example, the workshops for Week 2 concern lectures 1B & 2A; the workshops for Week 3 concern lectures 2B & 3A, etc.

Remember that workshop exercises are designed to be done in class with the assistance of a tutor. While you should prepare for these classes, you should not expect to be able to do all the exercises unaided. (If you want to try easier exercises, you can find them in the text book, many with answers supplied.)

At the end of each week answers to the exercises will be available on LMS.

Week 1: Reading Week. No workshops. Use this week to finish preliminary reading.

Week 2: Recognising Argument

Decide of each of the following, giving reasons for your answer where possible, whether it offers an argument or not. If it does, then identify the conclusion and any inference indicators.

a. Sir Harry Hotspur of Humblethwaite was a mighty person in Cumberland, and one who well understood of what nature were the duties, and of what sort the magnificence, which his position as a great English commoner required of him.

b. Jane thought it must be Sunday because she could hear church bells, but then she remembered that it was only Thursday and concluded that it must be Ascension Day.

c. Of course he's guilty. Didn't he admit that he threatened to kill her?

d. All 80 students who dined in college on Friday became ill during the night. None of the students who live in college but who did not dine in on that evening became ill, so the illness must have been food poisoning caused by something served for dinner in college on Friday.

e. In England under the blasphemy laws it is illegal to express disbelief in the Christian religion. It is also illegal to teach what Christ taught on the subject of non-resistance. Therefore, whoever wishes to avoid being a criminal must profess to agree with Christ's teachings but must avoid saying what that teaching was.

f. The reason I oppose the amendment is because of Section 2, which says 'The Federal Parliament shall have the power to enforce by appropriate legislation the provisions of this Article.' This section transfers power from the states to Canberra, and the states have lost too many powers already.

g. It is characteristic of Newtonian physics that it has to ascribe independent and real existence to space and time as well as to matter; for in Newton's law of motion the concept of acceleration appears. But in this theory, acceleration can only denote 'acceleration with respect to space'. Newton's space must thus be thought of as 'at rest', or at least as 'unaccelerated', in order that one can consider the acceleration, which appears in the law of motion, as being a magnitude with any meaning. Much the same holds with time, which of course likewise enters into the concept of acceleration. Newton himself and his most critical contemporaries felt it to be disturbing that one had to ascribe physical reality both to space itself as well as to its state of motion; but there was at that time no other alternative, if one wished to ascribe to mechanics a clear meaning.

Week 3: Standardising Arguments

Put letters and angle brackets in the following arguments, and underline all inference indicators. Then, using the letters you have assigned, rewrite the arguments in revised standard form, adding any omitted material in square brackets.

1. Beware; for I am fearless, and therefore powerful. (Mary Shelley, *Frankenstein*)

2. The world could not have been created three days before the sun. Either the concept of a day is not defined by reference to the sun, or there could not have been a world three days before there was a sun, since if a day is defined this way, there would be no way of measuring days before there was a sun. The concept of a day is defined as a period between one sunrise and the next; that means that it is defined by reference to the sun. Thus, it follows that at least one thing in the Bible is false.

3. The "total energy" of a physical system, a meaningful concept for flat space, is a meaningless concept if we allow for arbitrary space curvature. Since the space of the entire universe can curve, the total energy of the universe is thus simply not a meaningful concept. This conclusion, that the concepts of total energy and total energy conservation do not apply to the whole universe, is quite startling — but true. It implies that if we are to define *nothing* — the vacuum state — as it might apply to the whole universe, then we ought to look for features of the vacuum that do not use the concept of "total energy".

4. [*Background: The following argument comes from Part I Chapter 14 of the seventeenth century work* Leviathan, *by Thomas Hobbes.*]
Whensoever a man Transferreth his Right, or Renounceth it; it is either in consideration of some Right reciprocally transferred to himselfe; or for some other good he hopeth for thereby. For it is a voluntary act: and of the voluntary acts of every man, the object is some *Good to himselfe*. And therefore there be some Rights, which no man can be understood by any words, or other signes, to have abandoned, or transferred.

Week 4: Support and acceptability

1. Identify the conclusion(s) in each of the arguments which follow, and evaluate the degree of support which the premisses give to the conclusion(s). Give reasons for your answers.

a. Inasmuch as the defendant's footprints were found at the scene of the murder, and they would not be found there if he were not the murderer, we may conclude that the defendant must be the murderer.

b. If the state of Dalwhinnie develops the ability to launch, without harming itself, a surprise attack on the armed forces of its enemy Lagavulin, an attack that is so devastating that Lagavulin would be completely unable to retaliate, this would increase Dalwhinnian national security, because if Dalwhinnie develops such a first-strike capability, then Lagavulin will be more afraid of Dalwhinnian power.

c. Growing food for local consumption receives low priority from many governments of African nations. In 1983, drought devastated millet production in the northern provinces of Upper Volta, yet farmers in the south — with bumper surpluses of millet to sell — were discussing planting cotton instead. The reason is simple. The farmers know that cotton for export will quickly be collected and paid for by the government. Thus, cotton production has shot up over 20 times since independence while yields of sorghum and millet, the major food crops for the region, have stagnated.

The trend is typical for many African countries. In Mali, during the great drought between 1976 and 1982, while food production plummeted, cotton production increased by 400 per cent. During the drought of 1973–4 in Tanzania, sales of maize fell by a third while the output of tobacco continued to grow. Both crops need about the same amount of rain. The difference comes in inputs available and the incentives to growers. Sixty two per cent of money lent by the Tanzanian Rural Development Bank between 1978 and 1979 went for tobacco and only nineteen per cent for maize.

2. Give examples of the following:

a. Necessary and contingent truths (and falsehoods), both obvious and unobvious.

b. Explicit and implicit contradictions.

3. Devise a brief argument for a conclusion of your choice, and then describe two different contexts in which the argument is put forward, so that the same premiss is acceptable in one of the contexts but not in the other.

The next workshop topic involves a long argument; it is essential that you study the argument closely before the workshop.

Week 5: Evaluating a long argument

(You must study this argument closely beforehand.)

The following is a slightly adapted version of a real case of argument at length. (The numbers correspond merely to paragraphs in the original, to enable easy reference.) Evaluate the reasoning.

1. Every wave of education reform brings with it some good ideas and others that are attractive but not well thought out. The current reform wave is no exception.

2. One of the great virtues of the current wave is that it recognizes the classroom teacher as the key to a good education. (It's hard to believe that a few years ago an earlier generation of reformers pinned their hopes on the creation of "teacher-proof" educational materials.) Many of the recent reports list recommendations designed to retain our teaching force and attract new talent, but it is inevitable that any discussion of teacher quality will also deal with the question of how to get rid of incompetents.

3. All of us were at one time students, and while we were lucky enough to have one, two or three great teachers who inspired us, the chances are good that we had just as many teachers at the other end of the quality spectrum. Of course, there are occasions when a given teacher rated as incompetent by one student or principal would be deemed satisfactory or even outstanding by others. Still, teaching has its share of incompetents, as does every other field, and the reform wave has resurrected some old "cures".

4. Most popular is the idea that tenure should be abolished so that teachers or faculty thought to be incompetent by their supervisors may be dismissed without the protection of tenure — the right to know why the discharge is being sought and to challenge it before an impartial body that has the opportunity to hear both sides of the case. Somehow the public thinks that tenure gives teachers something that no one else enjoys — a special protection. It's just not so. What teachers have is no better than, and usually weaker and worse than, that enjoyed by other professionals — doctors, dentists, lawyers, pharmacists — and millions of workers in both the private and public sectors who can only be dismissed for "just cause".

5. But the real question is, would abolishing tenure lead to better teaching, better results in school? We don't have to abolish tenure in order to find out. Since at least two states do not have tenure — Mississippi and Texas — if tenure were the big stumbling block to improving education by easily removing incompetents, someone should be able to demonstrate the superiority of education in these two states. Anyone with the evidence ought to produce it. And if there is no evidence, it's time to put this idea back into mothballs.

6. A second idea is recertification. Even people who were really good five, ten, fifteen years ago get worn out — everyone knows that. Why require a "trial" to get someone out? Why not have all teaching certificates terminate after a specific period of time, such as five years? Most teachers' licences would be renewed after five years, but some would not be, thus avoiding messy disputes and arguments.

7. The idea is appealing, but why apply it only to teachers? Why not to dentists, doctors, lawyers, pharmacists? Any of these whose performance deteriorates is likely to do at least as much damage as a teacher. If recertification were imposed on all of these and other job categories (and I don't mean a *pro forma* recertification but one that really reviews the desirability of granting another five years), it would be hard to ask that teachers be exempt, but it's just as unfair to subject teachers to this when no one else's certification is up for review. Also, we'd be sending a message to many who are now in college: In addition to having low pay and low prestige, the profession they are preparing to enter can easily be taken away from them every five years. We'd thus be adding exceptional insecurity to the other disadvantages of the job.

8. Third is the idea of retesting teachers. I've been a strong advocate of giving teachers tests before they're hired and requiring high passing marks in the subject matter to be taught. But why give a pencil-and-paper test to a teacher after five or ten years when you can spend a few hours in his or her classroom and find out by direct observation whether the teacher knows the subject and knows how to teach it? Of course, knowledge in a given field is always expanding. Shouldn't teachers be tested again to make sure they're keeping up with the latest advances?

9. Again, the answer is yes — but only if they're not singled out. The knowledge explosion in medicine, dentistry and pharmacy is much greater than new developments in the teaching of Shakespeare, Dickens, irregular French verbs, high school algebra or fourth-grade arithmetic. Here, too, if we're going to retest teachers to protect the public, it's going to be hard to convince teachers that they and only they should be retested.

10. You can't blame teachers for feeling that they're being made scapegoats when these proposals are floated by officials who are trying to hire teachers at the lowest possible rates, who don't bother giving teachers a test before they're hired, and who don't follow a policy of granting tenure only to those who meet very high standards during their probationary period. (Albert Shanker)

Week 6: Devising a *reductio*

1. Construct an argument by *reductio ad absurdum* (in the narrow sense) that the following statement could not form part of a rational procedure for the conduct of meetings of the Philosophy Department:

At a Philosophy Department meeting, a motion is passed if at least 50% of those present vote in favour of it, and is not passed if at least 50% of those present vote against it.

2. Show by a *reductio* argument in the broad sense that it is impossible to move a rook (i.e. castle), using only normal rook moves, from one corner of a chessboard to the corner diagonally opposite so that it passes through every square exactly once. (Assume that starting on a corner square counts as passing through it.)